

Multiply rational expressions



1 Fully simplify the following expressions:

a $\frac{c}{p} \times \frac{f}{v}$

b $\frac{7x}{6} \times \frac{5y}{2}$

c $\frac{11s}{7t} \times \frac{3r}{5q}$

d $\frac{11}{5y} \times \frac{7}{2y}$

e $\frac{8u}{3v} \times \frac{2v}{7u}$

f $\frac{7x}{4y} \times \frac{7y}{10x}$

g $\frac{x^2}{5} \times \frac{15}{x}$

h $\frac{2n^2}{6} \times \frac{-2y^2}{3n}$

i $\frac{7a^2}{15b^2} \times \frac{10b}{11a}$

j $\frac{5y}{2} \times 6y^5$

k $\frac{m}{3} \times \frac{5m}{9} \times 81q$

l $\frac{n^2}{4} \times \frac{2n^4}{5} \times 20n$

2 Fully simplify the following expressions:

a $\frac{5x+8}{8xy^2} \times \frac{9xy}{25x+40}$

b $\frac{7x+8}{6xy^2} \times \frac{xy}{14x+16}$

c $\frac{p+7}{5} \times \frac{5p-2}{p^2+14p+49}$

d $\frac{8x+72}{x+2} \times \frac{3x^2+6x}{x^2+2x-63}$

e $\frac{4b(b+8)}{(b+8)^2} \times \frac{(b+8)(b-1)}{6b}$

f $\frac{4m(4m+3)}{(4m+3)^2} \times \frac{(4m+3)(m-3)}{6m}$

g $\frac{a^2-16}{a(a+4)} \times \frac{7a+28}{28(a-4)}$

h $\frac{x(x-8)}{x^2+16x+64} \times \frac{16(x+8)}{-2x+16}$

i $\frac{x^2-4}{36-x^2} \cdot \frac{x^2-8x+12}{8-12x+6x^2-x^3}$

j $\frac{9xy+18x}{x-9} \times \frac{xy-3x-9y+27}{4y^2+8y}$

Divide rational expressions

3 Fully simplify the following expressions:

a $\frac{7x}{6} \div \frac{4}{11y}$

b $\frac{m}{3} \div \frac{m}{2}$

c $\frac{5y}{9} \div \frac{4y}{11}$

d $\frac{33u}{10y} \div \frac{88q}{25y}$

e $\left(-\frac{7}{9}\right) \div \left(-\frac{14n}{27}\right)$

f $\frac{16u}{17v} \div (-14uv)$

g $\frac{u^2}{9} \div \frac{u}{6}$

h $\frac{45x^2yz}{10} \div \frac{9y^2}{5x}$

i $\frac{35t^2u^2}{12v^2w^2} \div \frac{35t^3u}{12w^2y^2}$

j $\frac{77x^5y^3}{9z} \div \frac{55x^8}{y^2z}$

4 Fully simplify the following expressions:



a $\frac{4k-5}{2} \div \frac{12k-15}{4}$

c $\frac{k-1}{20} \div \frac{k^2-10k+9}{4}$

e $\frac{3m^2+3m}{7myz} \div \frac{10m+10}{3mz}$

g $\frac{5x+5}{4(x+5)} \div \frac{7(x-9)}{2x^2-8x-90}$

i $\frac{x-2}{3x^2-30x+63} \div \frac{x^2-4}{5x^2-65x+210}$

b $\frac{k-1}{2} \div \frac{k^2-1}{6}$

d $\frac{3ab-30a}{7bc+49b} \div \frac{7ab-70a}{5bc+35b}$

f $\frac{6x(y+6)}{xy+6x+6y+36} \div \frac{36xy}{6x+36}$

h $\frac{(m+9)(m+5)}{20(m+8)} \div \frac{(m+5)(m+7)}{15(m+7)}$

j $\frac{4xz+4yz}{2x-3y} \div \frac{24xz-36yz}{4x^2-12xy+9y^2}$

5 Determine whether each of the following is equivalent to $\frac{7}{\frac{8}{x}}$:

a $\frac{7}{1} \times \frac{x}{8}$

b $\frac{7}{1} \times \frac{8}{x}$

c $\frac{7}{1} \div \frac{8}{x}$

d $\frac{1}{7} \div \frac{8}{x}$

6 Determine whether each of the following is equivalent to $\frac{\frac{x-5}{8}}{\frac{y+1}{9}}$:

a $\frac{x-5}{8} \div \frac{y+1}{9}$

b $\frac{y+1}{9} \div \frac{x-5}{8}$

c $\frac{x-5}{8} \times \frac{y+1}{9}$

d $\frac{x-5}{8} \times \frac{9}{y+1}$

7 Complete the following statements:

a $\frac{\frac{v}{3}}{\frac{4u}{3}} = \frac{\frac{(\frac{v}{3})}{\frac{4u}{3}}}{\frac{(\frac{4u}{3})}{\frac{3}{3}}} = \frac{\frac{(\frac{v}{3})}{\frac{4u}{3}}}{\frac{(\frac{4u}{3})}{1}}$

b $\frac{\frac{u}{4}}{\frac{v}{8}} = \frac{\frac{(\frac{u}{4})}{\frac{v}{8}}}{\frac{(\frac{v}{8})}{\frac{8}{8}}} = \frac{\frac{(\frac{u}{4})}{\frac{v}{8}}}{\frac{(\frac{v}{8})}{1}}$

c $\frac{\frac{y}{2}}{\frac{3x}{2}} = \frac{\frac{(\frac{y}{2})}{\frac{3x}{2}}}{\frac{(\frac{3x}{2})}{\frac{2}{2}}} = \frac{\frac{(\frac{y}{2})}{\frac{3x}{2}}}{\frac{(\frac{3x}{2})}{1}}$

d $\frac{\frac{4}{n}}{\frac{5}{n^2}} = \frac{\frac{(\frac{4}{n})}{\frac{5}{n^2}}}{\frac{(\frac{5}{n^2})}{\frac{n^2}{n^2}}} = \frac{\frac{(\frac{4}{n})}{\frac{5}{n^2}}}{\frac{(\frac{5}{n^2})}{1}}$

8 Fully simplify the following expressions:



a
$$\frac{\frac{1}{2}}{\frac{3}{4}}$$

b
$$\frac{\frac{9x}{4}}{\frac{3x}{2}}$$

c
$$\frac{\frac{y^2}{6}}{\frac{y}{18}}$$

d
$$\frac{\frac{3}{10x^2}}{\frac{15}{16x^3}}$$

e
$$\frac{\frac{4y-6}{10}}{\frac{6y-9}{5}}$$

f
$$\frac{\frac{9y-12}{15}}{\frac{12y-16}{5}}$$

g
$$\frac{\frac{8x-6}{3x^2}}{\frac{4x-3}{9x}}$$

h
$$\frac{\frac{4x-2}{4x^2}}{\frac{2x-1}{8x}}$$

Applications

9 The product of two fractions is 1. If one of the fractions is $\frac{5x}{9y}$, what is the other fraction?

10 A rectangle has an area of $\frac{5x^3y^4}{3pq}$ units² and a length of $\frac{4xy}{p}$ units. Find an expression for the width of the rectangle.